Axler Linear Algebra

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Contents

I plan to mostly take notes on paper, but I will a lot of the exercises here because I need to practice writing LaTeX quickly and it's also a nicer format for other people to read.

1 Vector Spaces

1.A $\mathbb{R}^n$ and $\mathbb{C}^n$

These exercises are really easy but I am always scared that my proofs of these are not rigorous enough....

I do odds.

1 Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Let $\alpha = a + bi$ and $\beta = c + di$ where $a, b, c, d \in \mathbb{R}$.

Then we have

$$\begin{aligned} \alpha + \beta &= \\ a + bi + c + di &= \\ (a + c) + (b + d)i &= \end{aligned}$$

Because addition is commutative under the reals, we can swap the terms.

$$\begin{aligned} (c + a) + (d + b)i &= \\ c + di + a + bi &= \\ \beta + \alpha \end{aligned}$$

$\square$

3 Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Distribute, multiply, then rearrange terms using the communitivity of the reals to get the answer. It is quite tedious, so I won't do it here.

5 Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Let $\alpha = a + bi$. Then, let $\beta = -a - bi$. We can observe that

$$\begin{aligned} \alpha + \beta &= \\ a + bi - a - bi &= \\ (a - a) + i(b - b) &= 0 \end{aligned}$$

$\square$

7 Show that $\frac{-1+\sqrt{3}i}{2}$ is a cube root of 1 (meaning that its cube equals 1)

Again, this is a tedious problem where you just multiply it out. Of course, I can represent this number as $e^{\frac{\pi}{3}i}$ but that's cheating.

9 Find $x \in \mathbf{R}^4$ such that

$(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$

Proof. This is vector addition, so we can apply the definition.

$$\begin{aligned} (4, -3, 1, 7) + 2x &= (5, 9, -6, 8) \\ (4 + 2x_1, -3 + 2x_2, 1 + 2x_3, 7 + 2x_4) &= (5, 0, -6, 8) \end{aligned}$$

For each term in the list, the number must hold, so in essence we get four different equations:

$$\begin{aligned} 4 + 2x_1 &= 5 \\ -3 + 2x_2 &= 0 \\ 1 + 2x_3 &= -6 \\ 7 + 2x_4 &= 8 \end{aligned}$$

Each of these linear equations yields only one solution, so the only solution for the equation is $\boxed{x = (\frac{1}{2}, -\frac{3}{2}, -\frac{7}{2}, \frac{1}{2})}$ □

1.B Definition of a Vector Space

This one introduces the concept of a vector space in an abstract manner and gives a few unorthodox examples.

1 Prove that $-(-v) = v$ for every $v \in V$

Proof. $-(-v)$ is the additive inverse of $-v$. Therefore,

$$\begin{aligned} -(-v) + (-v) &= 0 \\ -(-v) + (-v + v) &= v \\ -(-v) + 0 &= v \\ -(-v) &= v \end{aligned}$$

As desired. □

3 Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

$$\begin{aligned} v + 3x &= w \\ 3x &= w - v \\ x &= \frac{1}{3}(w - v) \end{aligned}$$

Since that right side is a unique vector, x is a unique vector. You invoke the properties and stuff for this, but I'm too lazy.

5 Show that in the definition of a vector space (1.20), the additive inverse condition can be replaced with the condition that

$$0v = 0 \text{ for all } v \in V.$$

Here the 0 on the left side is the number 0, and the 0 on the right side is the additive identity of V.

Proof. We can show that given this condition, every element $v \in V$ has an additive inverse.

$$\begin{aligned} 0v &= 0 \\ (1 + -1)v &= 0 \end{aligned}$$

Note that even though we don't know whether *vector* addition has an additive inverse the 0 on the left size is just a scalar, so we can still say there is an additive inverse.

Then distribute and apply the multiplicative identity axiom…

$$\begin{aligned} 1v + (-1)v &= 0 \\ v + (-1)v &= 0 \end{aligned}$$

Therefore $(-1)v$ is the additive inverse for all $v \in V$. □

7 Suppose S is a nonempty set. Let V^S denote the set of functions from S to V. Define a natural addition and scalar multiplication on V^S , and show that V^S is a vector space with these definitions.

Proof. Like with F^S, we can sort of "piggyback" off the properties of V to maintain the properties of a vector space in V^S.

Define addition as

$$(f + g)(x) = f(x) + g(x)$$

Let the additive identity be $0(x) = 0$ for all $x \in S$.

$$(f + 0)(x) = f(x) + 0(x) = f(x) + 0 = f(x)$$

For all $f, g \in V^S$. Because $f(x)$ is a vector, it has an inverse (namely $-f(x)$). Therefore, we can define the additive inverse as

$$(-f)(x) = -f(x)$$

$$(f + (-f))(x) = f(x) + (-f(x)) = 0 = 0(x)$$

Communativity:

$$(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$$

We can swap $f(x)$ and $g(x)$ because V is communative.

Associativity:

$$((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x)$$

Again, we use the properties of the vector space V to help us out here.

Scalar multiplication

For all $\lambda \in \mathbb{F}, v \in V^S$,

$$(\lambda v)(x) = \lambda v(x)$$

Multiplicative Identity

1 is the multiplicative identity.

$$(1v)(x) = 1v(x) = v(x)$$

Since 1 is the multiplicative identity of V.

Distributive property

$$(a(u + v))(x) = a(u + v)(x) = a(u(x) + v(x)) = au(x) + av(x) = (au + av)(x)$$

Because of the distributive property of V.

$$((a + b)v)(x) = (a + b)v(x) = av(x) + bv(x) = (av + bv)(x)$$

For the same reason.

Thus, we have proved all the properties of vector spaces on V^S, therefore V^S is a vector space.

Closure

Because V is closed under addition and scalar multiplication, and all our operations are defined using these 2 operations, V^S is closed.

$\square$